DSC 40B - Discussion 02

Problem 1.

Are the following asymptotic bounds true?

1. $n \log n = O(n^2)$
2. $\left(\frac{1}{n}\right)^3 = O(\frac{1}{n})^4$
3. $n^{\frac{1}{2}} = \Omega(n^{\frac{1}{3}})$
4. $e^n = O(n!)$
5. $\log n = \Omega(\sqrt{n})$

Problem 2.

a) Let $f(n) = 12log_2(3^{n^2-2n} + 2^{logn} - 10n^2 - log_3 n)$. Which of the following asymptotic bounds on f is true?

1. $f(n) = \Omega(1)$
2. $f(n) = O(1)$
3. $f(n) = O(\log n)$
4. $f(n) = \Omega(\log n)$
5. $f(n) = \Theta(\log n)$
6. $f(n) = O(n)$
7. $f(n) = O(e^n)$
8. $f(n) = \Theta(n)$
9. $f(n) = \Theta(n^2)$

b) What is the best case time complexity of the following function?

```
def foo(arr):
''' arr is a sorted array of size n'''
        i = 0
        j = len(arr) - 1

        while i < j:
                current_sum = arr[i] + arr[j]

                if current_sum == 5:
                        return sum(arr)
                elif current_sum < 5:
                        i += 1
                else:
                        j -= 1

        return False
```

Please also review problems 3 and 4 from discussion 1. They will help towards homework 2. In discussion on Friday, we will go over some of the problems from Homework 1 and 2.